

前情提要

(Previously on Preliminary)

經濟學是 使人更理智的一門學問

- 可見得人並不是隨時都是理智的
- 理智的行為是比較好分析的，唯有先明白理智行為為何，才有辦法凸顯人的不理性行為，什麼時候不理性及不理性的程度

我們要探討的理性經濟行為是：

(The rational behavior of human beings we would like to discuss is:)

Lecture 2: 需求、供給還有均衡

(Demand, Supply and the Equilibrium of Them)

而需求與供給的背後，是資源的配置

(It's a kind of allocation of scarce society's sources behind the demand and supply)

Outline

- 2-1: 個人(家庭)、公司(廠商)；要素市場、產品市場；政府的角色、市場機能
- 2-2:
 - 2-2.1: 需求量、需求曲線、需求法則
 - 2-2.2: 影響需求量的因素
- 2-3:
 - 2-3.1: 供給量、供給曲線、供給法則
 - 2-3.2: 影響供給量的因素
- 2-4: 從個人的供需曲線到市場的供需曲線
- 2-5:
 - 2-5.1: 供需均衡與均衡之外
 - 2-5.2: 市場機能與資源配置
- 2-6: 供需的另類詮釋

兩種經濟個體(two kinds of economic entities)

[1] 個人、家庭 (individuals, households)

- 擁有生產要素，將生產要素賣給廠商，向廠商買商品；
是要素供給者、商品需求者

[They own production factors and will sell them to the vendors. In other words, they are suppliers of production factors and demander of commodities]

[2] 廠商 (vendors)

- 向家庭購買生產要素，生產商品，賣給家庭；
是要素需求者、商品供給者

[They buy production factors from household to produce commodities and sell them back to earn profits. In other words, they are demanders of production factors and suppliers of commodities]

兩種市場(Two kinds of markets)

[1] 要素市場 (the market of production factors)

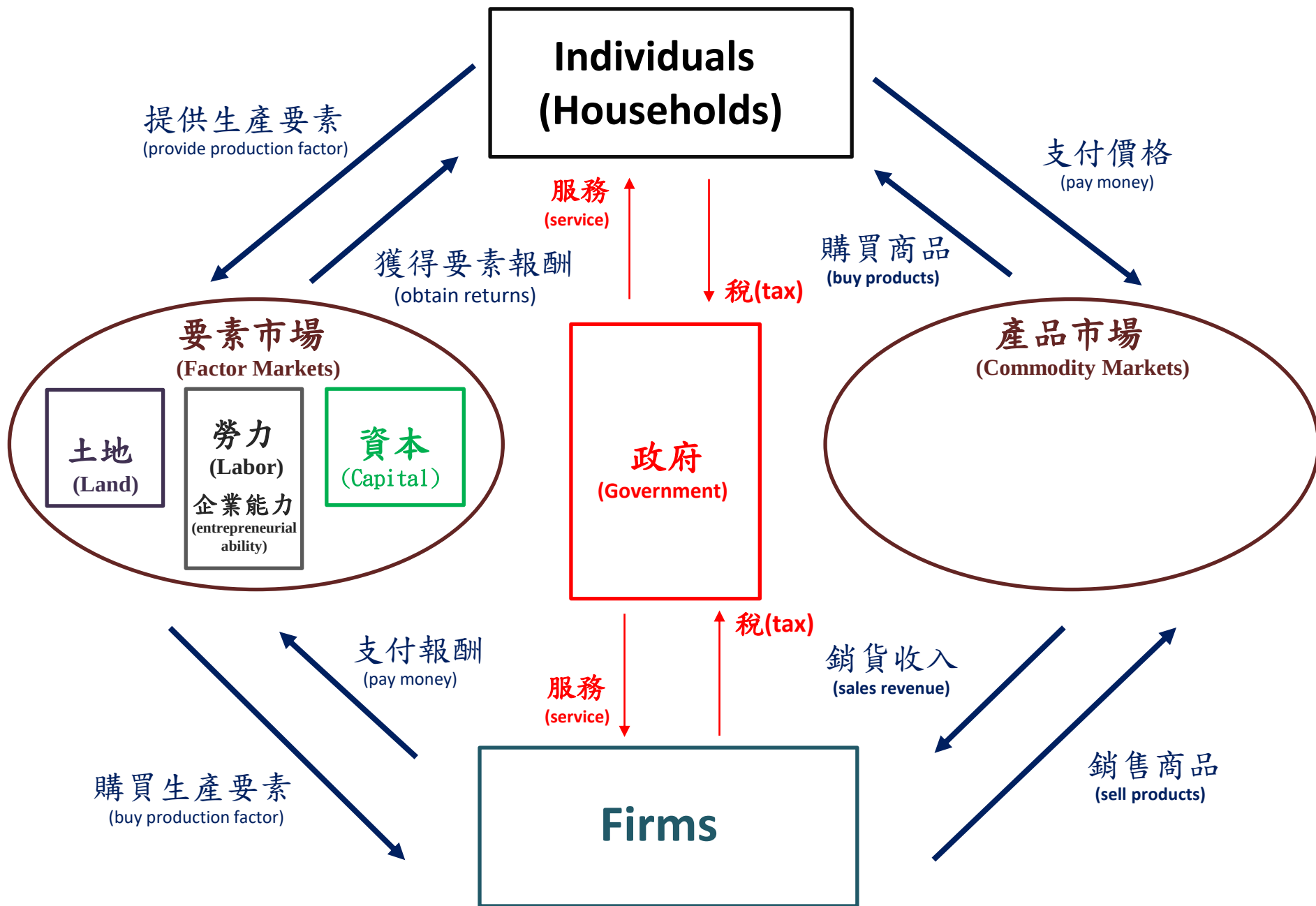
- 買賣生產要素之所在；有四種要素：土地、資本、勞動
企業能力

[The places that trade production factors. There are four kinds of production factors: land, capital, labor, and entrepreneurial ability.]

[2] 產品市場 (the market of commodities)

- 買賣商品之所在，交易標的為各種可供消費的商品

[The places that trade commodities.]



Markets and Competition (1)

- **Market**

- A group of buyers and sellers of a particular good or service
- Buyers as a group
 - Determine the demand for the product
- Sellers as a group
 - Determine the supply of the product

Markets and Competition (2)

- **Competitive market**
 - Many buyers and many sellers, each has a negligible impact on market price
- **Perfectly competitive market**
 - Buyers and sellers are so numerous that no one can affect the market price, “price takers
- Hereinafter we assume that all of the buyers and sellers are price takers

Demand

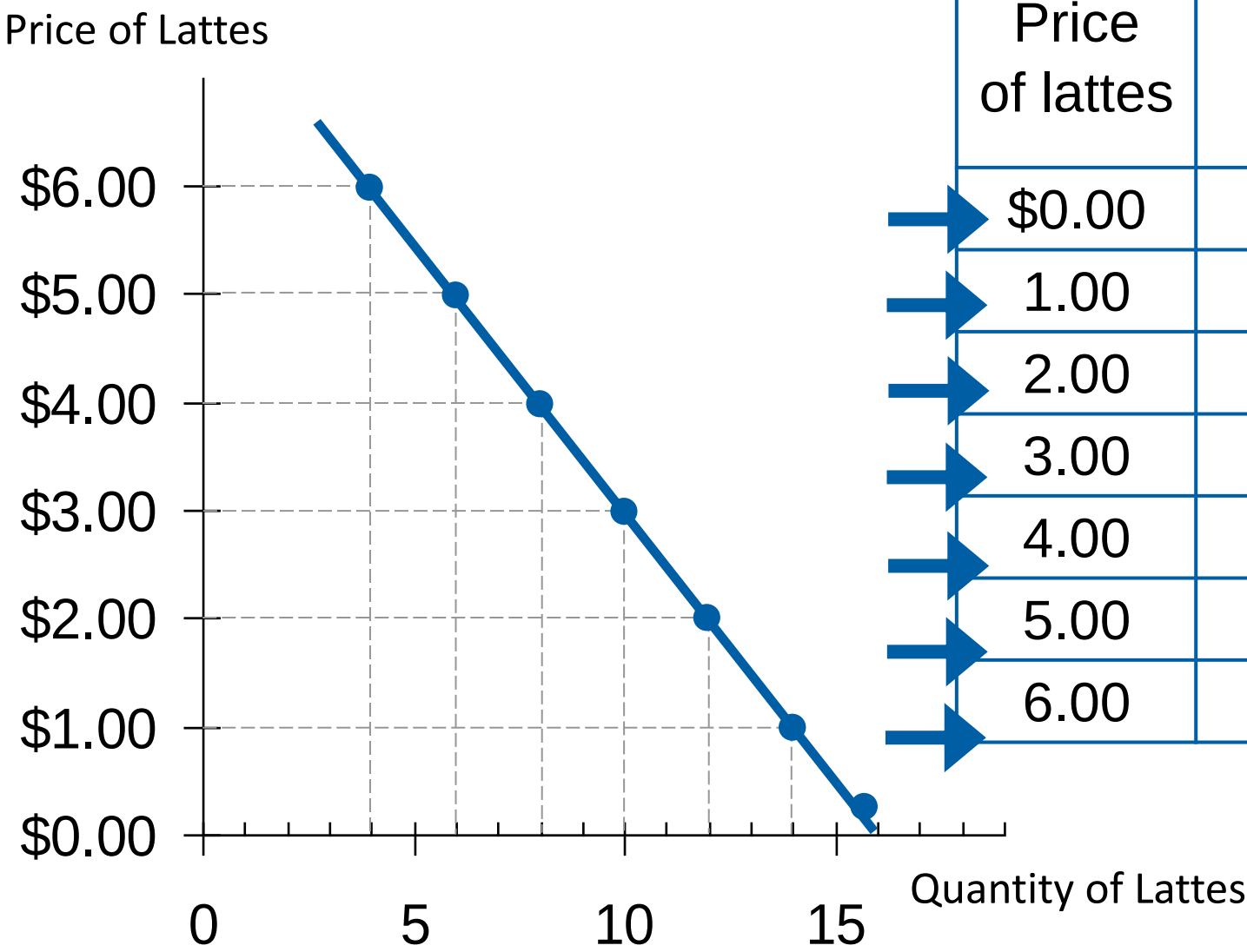
- **Quantity demanded**
 - Amount of a good that buyers are willing and able to purchase
- **Law of demand**
 - Other things equal
(e.g., given buyers' preference, income,...,etc.)
 - When the price of a good rises, the quantity demanded of the good falls
 - When the price falls, the quantity demanded rises

Sam's Demand Schedule

- Demand schedule:
 - A table, shows the relationship between the price of a good and the quantity demanded
 - Example: Sam's demand for lattes
 - Notice that Sam's preferences obey the law of demand.

Price of lattes	Quantity of lattes demanded
\$0.00	16
1.00	14
2.00	12
3.00	10
4.00	8
5.00	6
6.00	4

Sam's Demand Schedule and Demand Curve



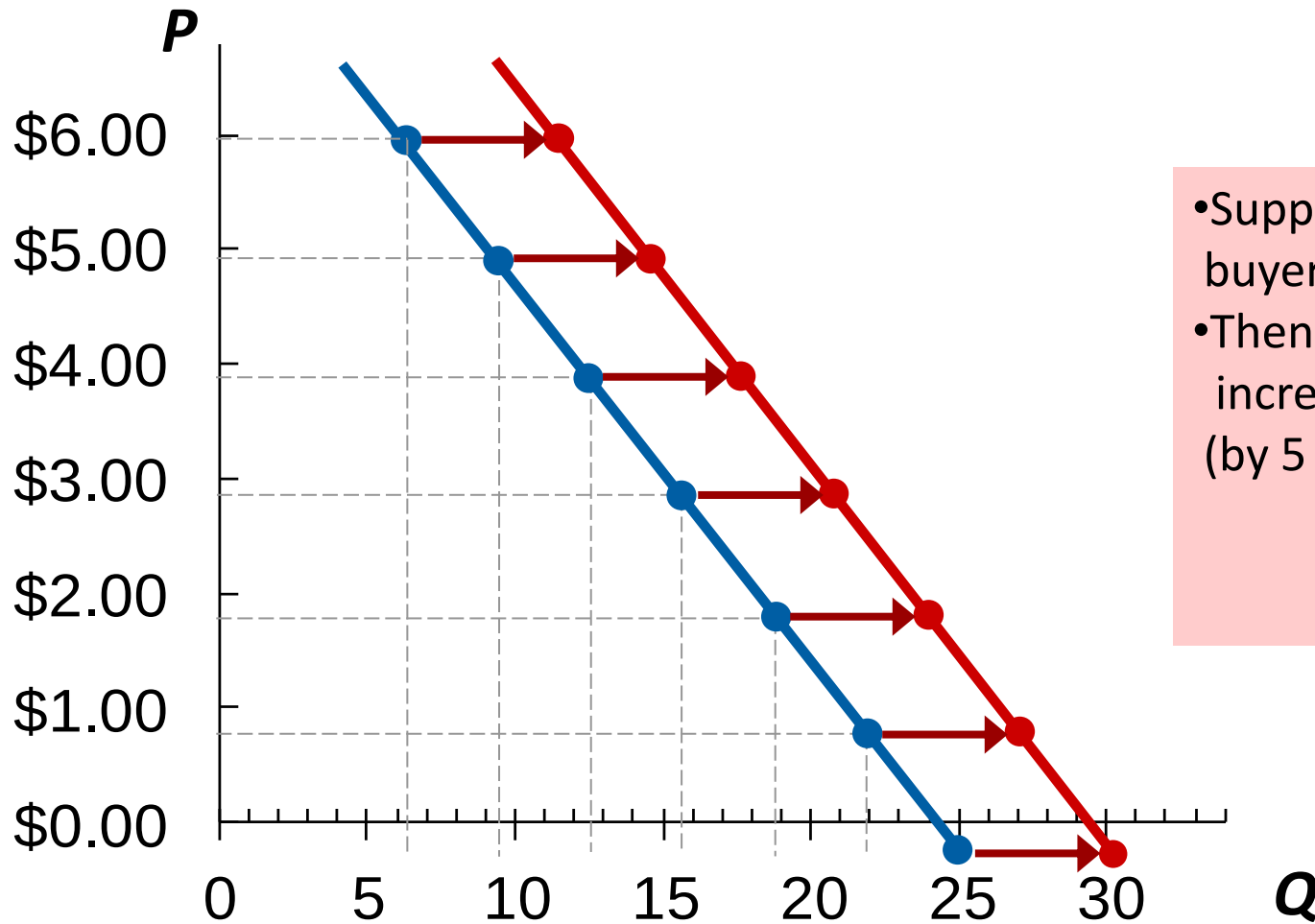
- **The demand curve**
 - Shows how price affects quantity demanded, other things being equal
- **These “other things” are non-price determinants of demand**
 - Things that determine buyers’ demand for a good, other than the good’s price
- Changes in them shift the demand curve.

Demand Curve Shifters

(1) Number of buyers

- Increase in number of buyers
 - Increases quantity demanded at each price
 - Shifts demand curve to the right
- Decrease in number of buyers
 - Decreases quantity demanded at each price
 - Shifts demand curve to the left

(1) Demand Curve Shifters: Number of Buyers



- Suppose the number of buyers increases.
- Then, at each P , Q^d will increase.
(by 5 in this example).

Demand Curve Shifters

(2)-1 Prices of related goods, substitutes

- Two goods are substitutes if
 - An increase in the price of one leads to an increase in the demand for the other
- **Example:** pizza and hamburgers
 - An increase in the price of pizza increases demand for hamburgers, shifting hamburger demand curve to the right
- **Other examples:**
 - Coke and Pepsi, laptops and tablets, music CDs and music downloads

(2)-2 Prices of related goods, complements

- Two goods are complements if
 - An increase in the price of one leads to a decrease in the demand for the other
- **Example:** computers and software
 - If price of computers rises, people buy fewer computers, and therefore less software; Software demand curve shifts left
- **Other examples:**
 - College tuition and textbooks, bagels and cream cheese, eggs and bacon

Demand Curve Shifters

(3) Income

- Normal good, other things constant
 - An increase in income leads to an increase in demand:
Shift the demand curve to the right
- Inferior good, other things constant
 - An increase in income leads to a decrease in demand:
Shift the demand curve to the left



Cathode ray tube(CRT) screen

vs.



LCD screen

Demand Curve Shifters

(4) Tastes

- Anything that causes a shift in tastes toward a good will increase demand for that good and shift its demand curve to the right

– Example:

- The Atkins diet (無澱粉飲食法) became popular in the '90s, caused an increase in demand for eggs, shifted the egg demand curve to the right

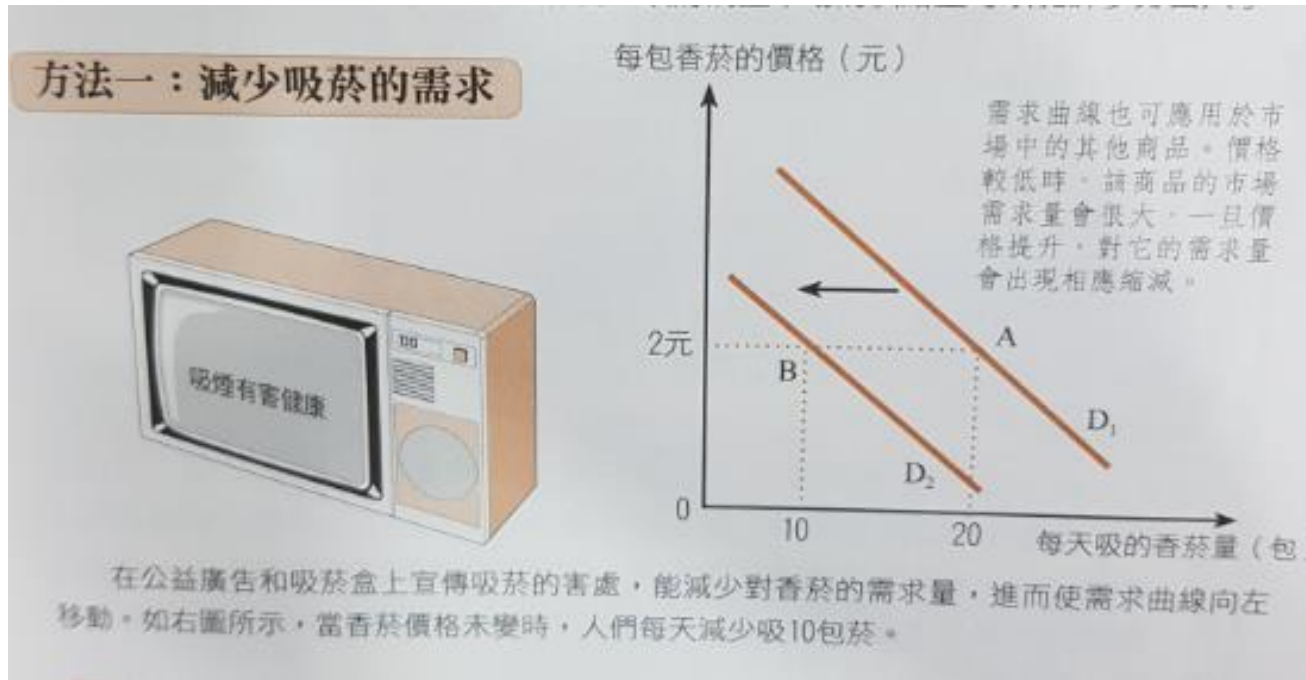
Demand Curve Shifters

(5) Expectations about the future

- Expect an increase in income, increase in current demand
- Expect higher prices, increase in current demand
- **Example:**
 - If people expect their incomes to rise, their demand curve for meals at expensive restaurants may increase now

國家降低香菸消費的方法 (Case Study)

(The way to suppress the consumption of cigarette from government point of view)



利用廣告或警語來改變消費者的偏好，使整個香菸的需求曲線左移，降低香菸的需求量
[Use advertisement or warnings to change buyers' preference. That could shift the demand curve to left side due to the decrease in the demand for the cigarettes.]

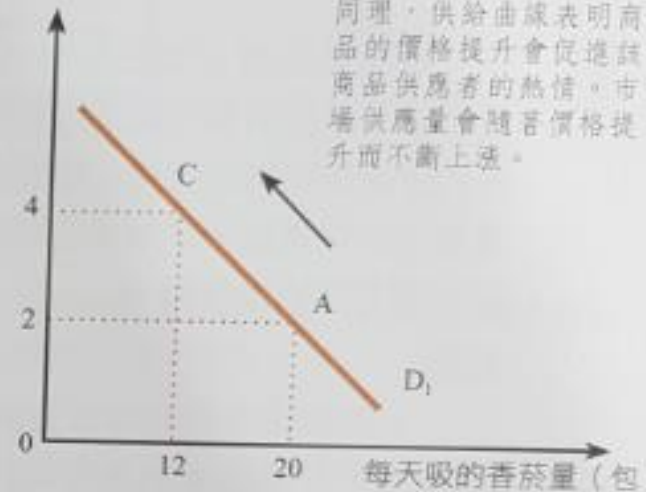
方法二：提高香菸的價格



徵稅

香菸價格提高

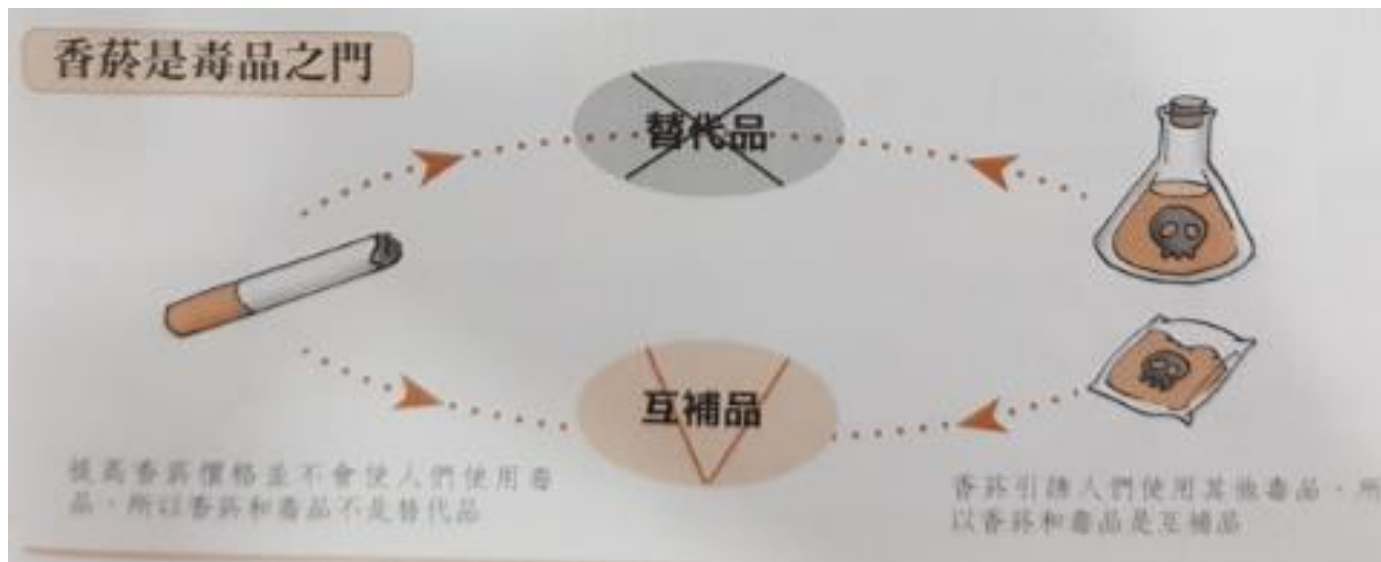
每包香菸的價格（元）



政府對菸草行業徵收香菸稅，促使菸草商提高了香菸價格。如右圖所示，需求曲線沒變，但人們的消費點向上移——在價格從20元上升到40元時，需求量從20包減少到12包。

利用課稅的手段使香菸價格提高，以降低香菸消費者的需求量

[Or tax the cigarette suppliers to force them push the price up. Higher cigarette price could suppress the buyers' demand for cigarette.]



(1) 香菸與毒品不是替代品，所以政府降低香菸的需求量並不會提高毒品的需求量

[Cigarettes and drugs are not substitutes. Thus, suppressing the demand for cigarettes does not increasing the demand for drugs]

(2) 香菸氾濫，很可能也同時提高毒品的需求量，所以香菸與毒品是互補品，有鑑於此，政府更應該打壓香菸的需求量

[High supply of cigarettes possibly pushes the demand for drugs up. Thus, cigarettes and drugs are complements. As a result, government needs suppress the demand for cigarettes]

Summary: Variables that Influence Buyers

Variable	A Change in This Variable . . .
Price of the good itself	Represents a movement along the demand curve
Income	Shifts the demand curve
Prices of related goods	Shifts the demand curve
Tastes	Shifts the demand curve
Expectations	Shifts the demand curve
Number of buyers	Shifts the demand curve

Supply

(See Mankiw, "Principles of Economics", 8th edition)

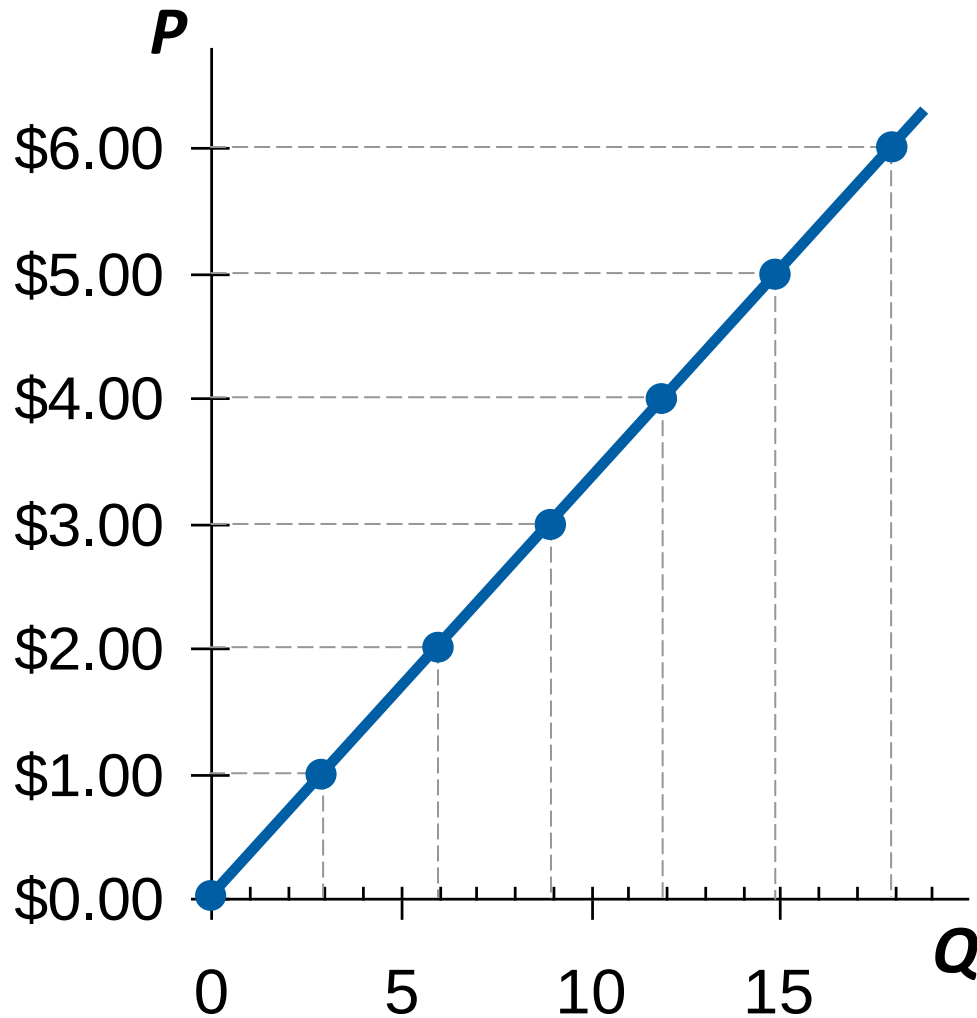
- **Quantity supplied**
 - Amount of a good sellers are willing and able to sell
- **Law of supply**
 - Other things equal
 - When the price of a good rises, the quantity supplied of the good rises
 - When the price falls, the quantity supplied falls

Starbucks' Supply Schedule

- Supply schedule:
 - A table, shows the relationship between the price of a good and the quantity supplied.
 - Example: Starbucks' supply of lattes
 - Notice that Starbucks' supply schedule obeys the law of supply

Price of lattes	Quantity of lattes supplied
\$0.00	0
1.00	3
2.00	6
3.00	9
4.00	12
5.00	15
6.00	18

Starbucks' Supply Schedule and Supply Curve



	Price of lattes	Quantity of lattes supplied
→	\$0.00	0
→	1.00	3
→	2.00	6
→	3.00	9
→	4.00	12
→	5.00	15
→	6.00	18

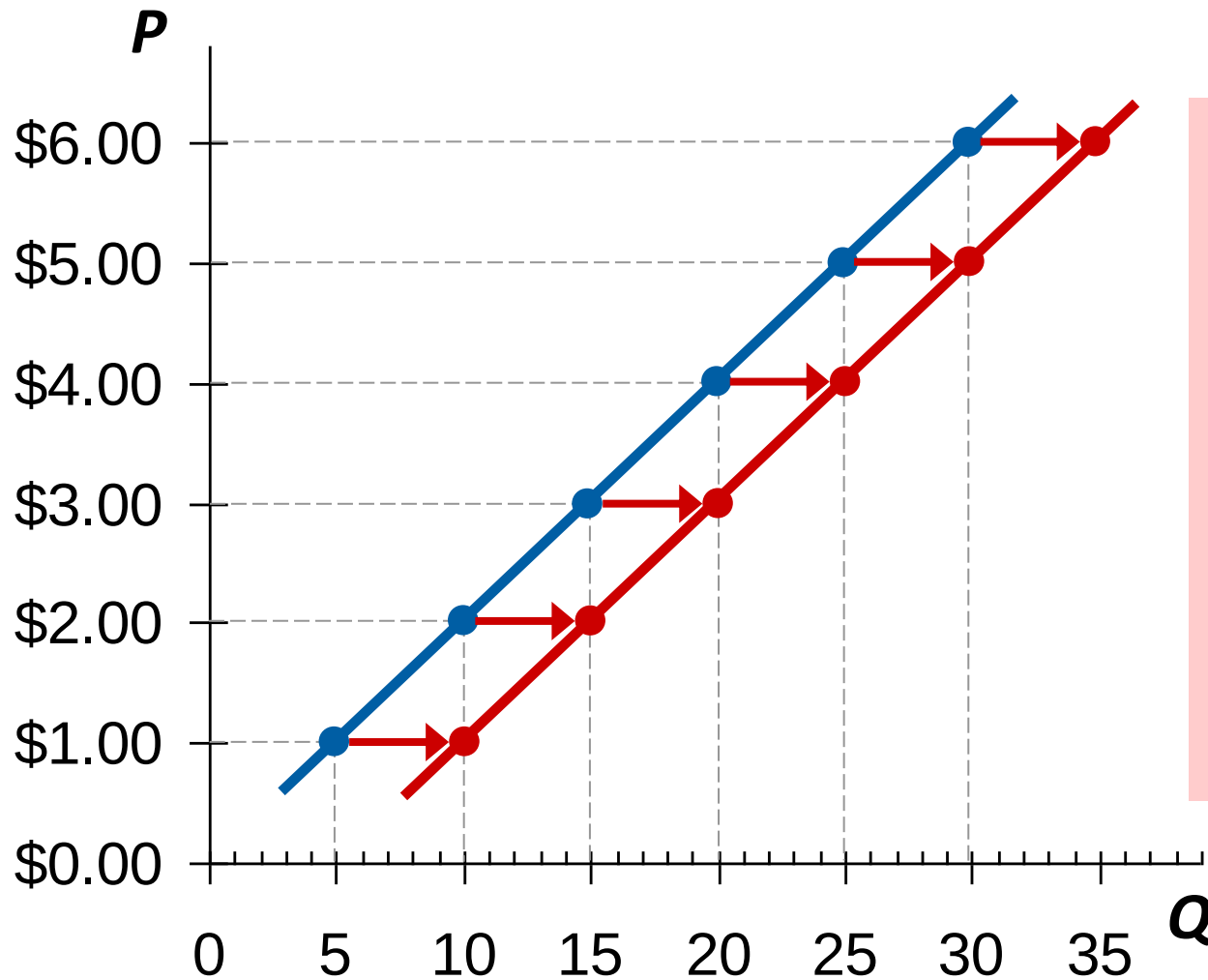
- **The supply curve**
 - Shows how price affects quantity supplied, other things being equal
- **These “other things” are non-price determinants of supply**
 - Things that determine sellers’ supply of a good, other than the good’s price
- Changes in them shift the supply curve.

Supply Curve Shifters

(1) Input prices

- Supply is negatively related to prices of inputs
- **Examples of input prices:**
 - wages, prices of raw materials
- A fall in input prices makes production more profitable at each output price
 - Firms supply a larger quantity at each price
 - The Supply curve shifts to the right

Supply Curve Shifters: Input Prices



- Suppose the price of milk falls.
- At each price, the quantity of lattes supplied will increase (by 5 in this example).

(2) Technology

- Determines how much inputs are required to produce a unit of output
- A cost-saving technological improvement has the same effect as a fall in input prices, shifts supply curve to the right

(3) Number of sellers

- An increase in the number of sellers (firms or vendors)
 - Increases the quantity supplied at each price
 - Shifts supply curve to the right

Supply Curve Shifters

(4) Expectations about future

- **Example:** Events in the Middle East lead to expectations of higher oil prices
 - Owners of Texas oilfields reduce supply now, save some inventory to sell later at the higher price
 - Supply curve shifts left

Summary: Variables That Influence Sellers

Variable	A Change in This Variable . . .
Price of the good itself	Represents a movement along the supply curve
Input prices	Shifts the supply curve
Technology	Shifts the supply curve
Expectations	Shifts the supply curve
Number of sellers	Shifts the supply curve

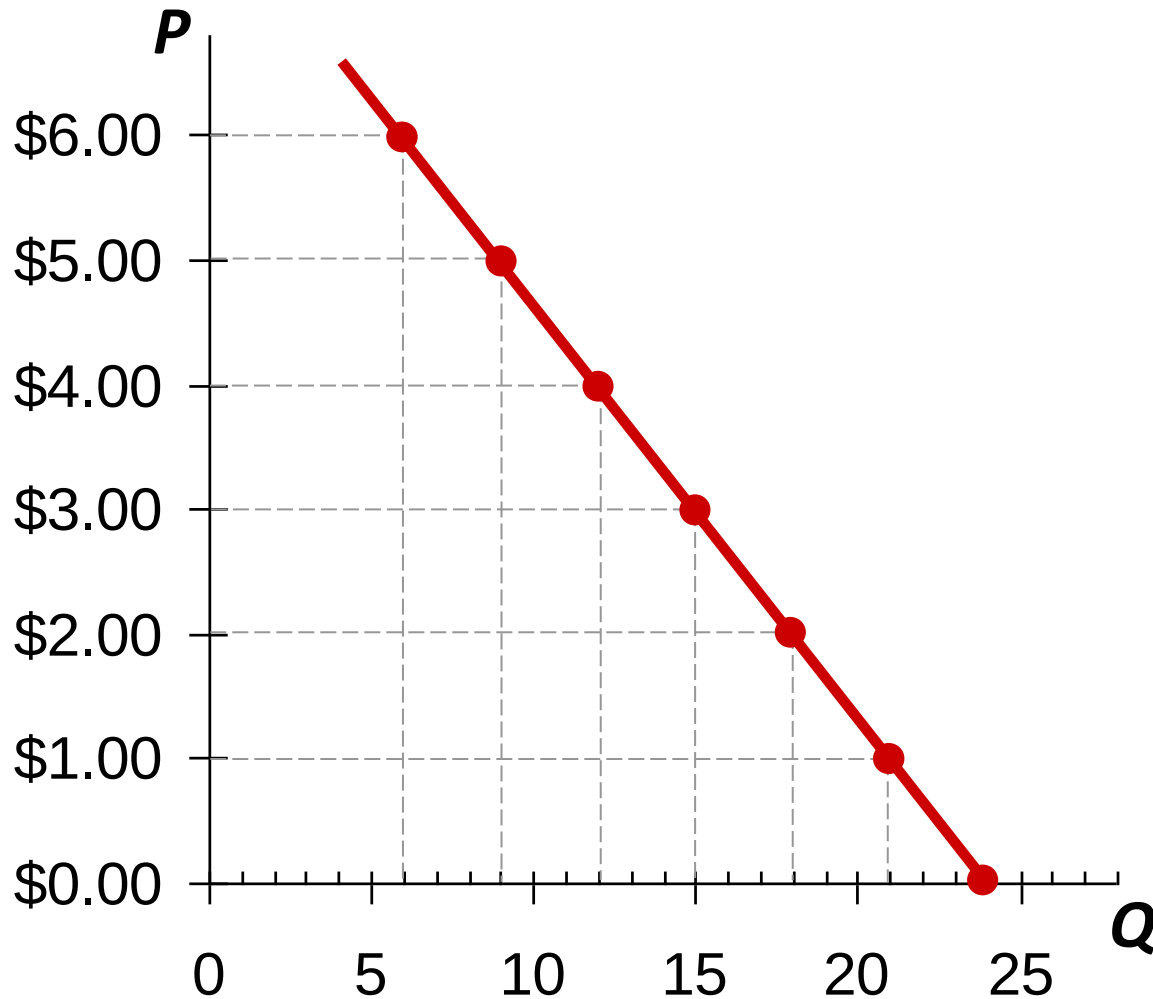
Market Demand vs. Individual Demand

- **Market demand**
 - Sum of all individual demands for a good or service
 - Market demand curve: sum the individual demand curves horizontally
 - To find the total quantity demanded at any price, we add the individual quantities

- Suppose Sam and Dean are the only two buyers in the market for lattes. (Q^d = quantity demanded)

Price	Sam's Q^d		Dean's Q^d		Market Q^d
\$0.00	16	+	8	=	24
1.00	14	+	7	=	21
2.00	12	+	6	=	18
3.00	10	+	5	=	15
4.00	8	+	4	=	12
5.00	6	+	3	=	9
6.00	4	+	2	=	6

The Market Demand Curve for Lattes



P	Q^d (Market)
\$0.00	24
1.00	21
2.00	18
3.00	15
4.00	12
5.00	9
6.00	6

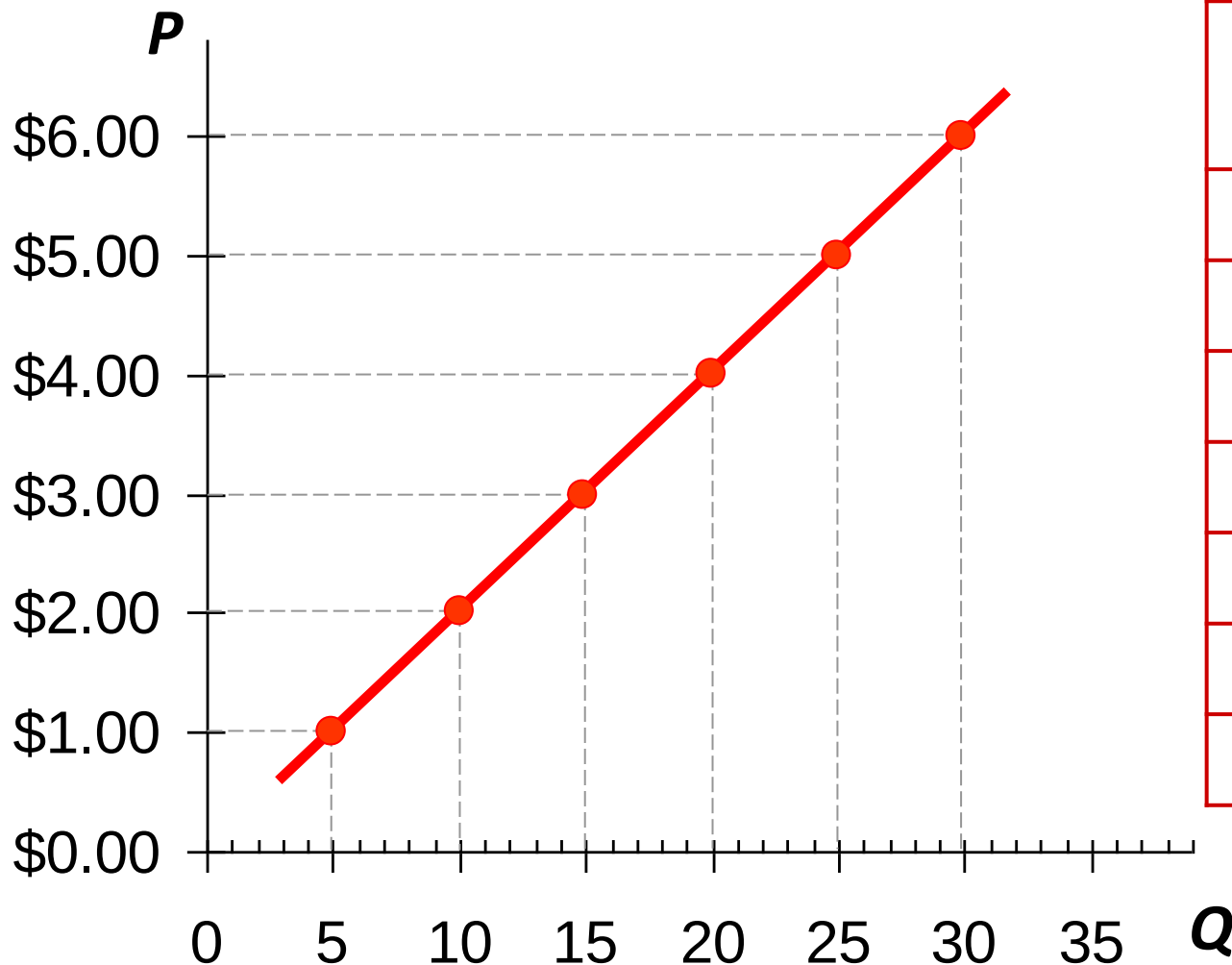
Market Supply vs. Individual Supply

- **Market supply**
 - Sum of the supplies of all sellers of a good or service
 - Market supply curve: sum of individual supply curves horizontally
 - To find the total quantity supplied at any price, we add the individual quantities

- Suppose Starbucks and Peet's are the only two sellers in this market. (Q^s = quantity supplied)

Price	Q^s Starbucks		Q^s Peet's		Market Q^s
\$0.00	0	+	0	=	0
1.00	3	+	2	=	5
2.00	6	+	4	=	10
3.00	9	+	6	=	15
4.00	12	+	8	=	20
5.00	15	+	10	=	25
6.00	18	+	12	=	30

The Market Supply Curve

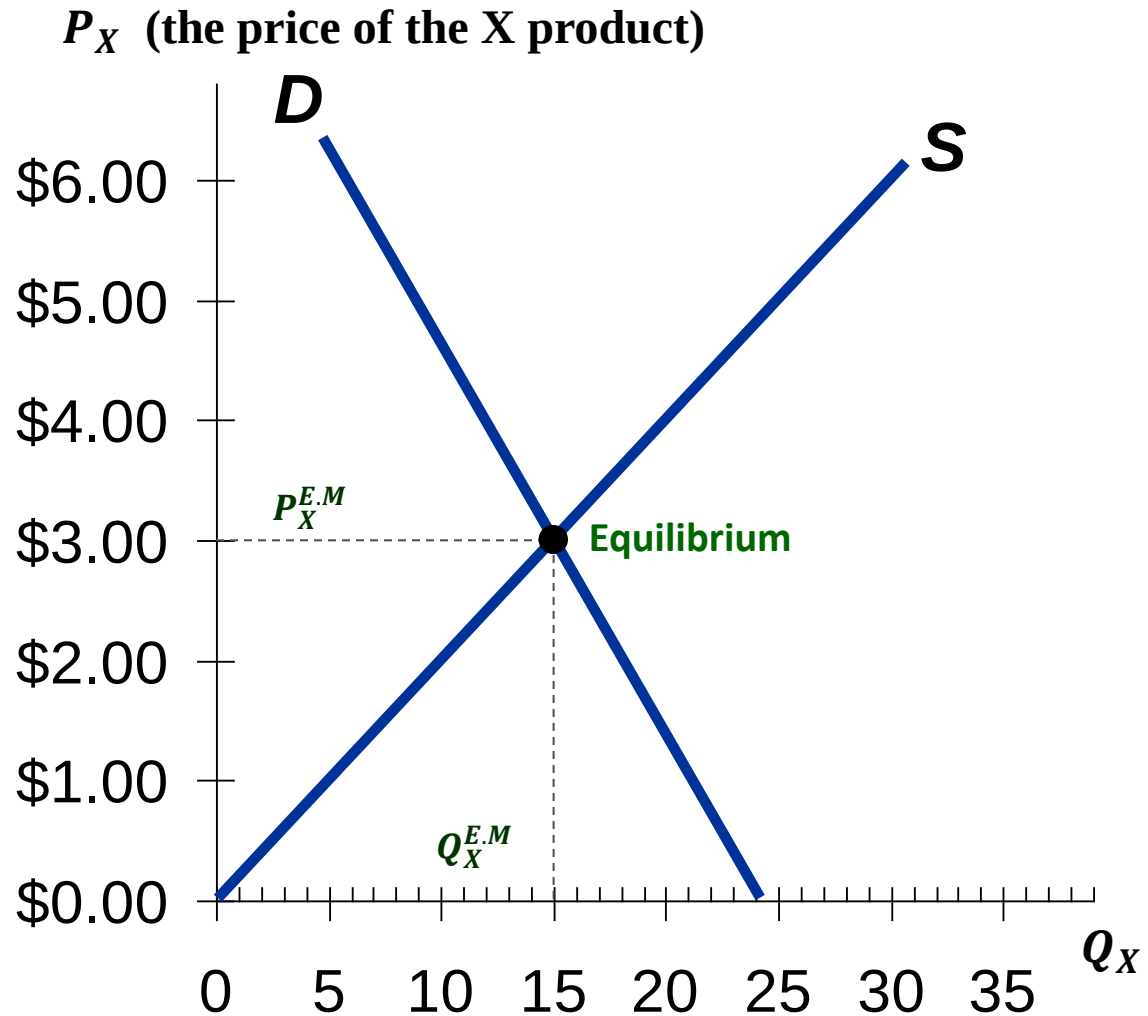


P	Q^s (Market)
\$0.00	0
1.00	5
2.00	10
3.00	15
4.00	20
5.00	25
6.00	30

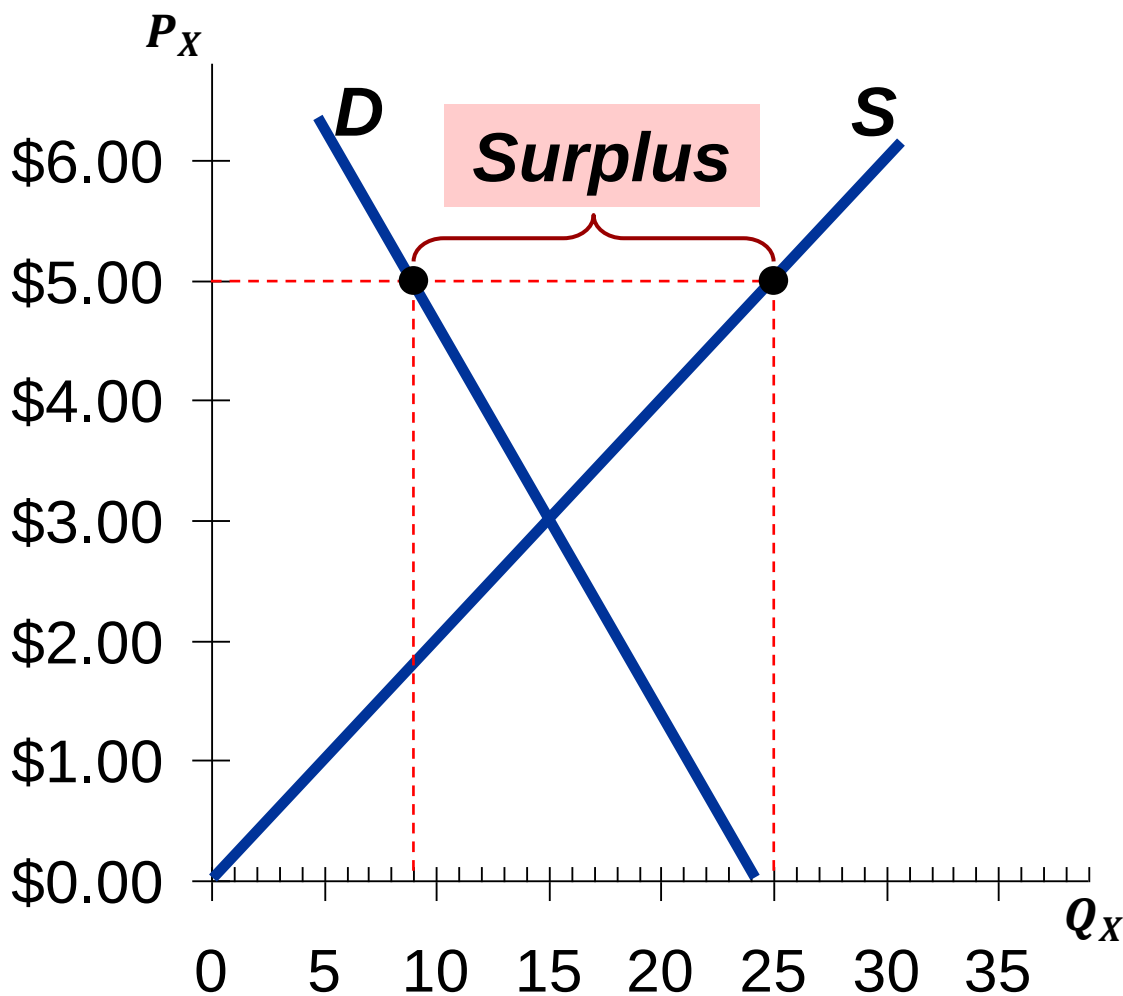
Supply and Demand Together

- **Equilibrium:**

Price has reached the level where quantity supplied equals quantity demanded



Markets Not in Equilibrium: **Surplus**

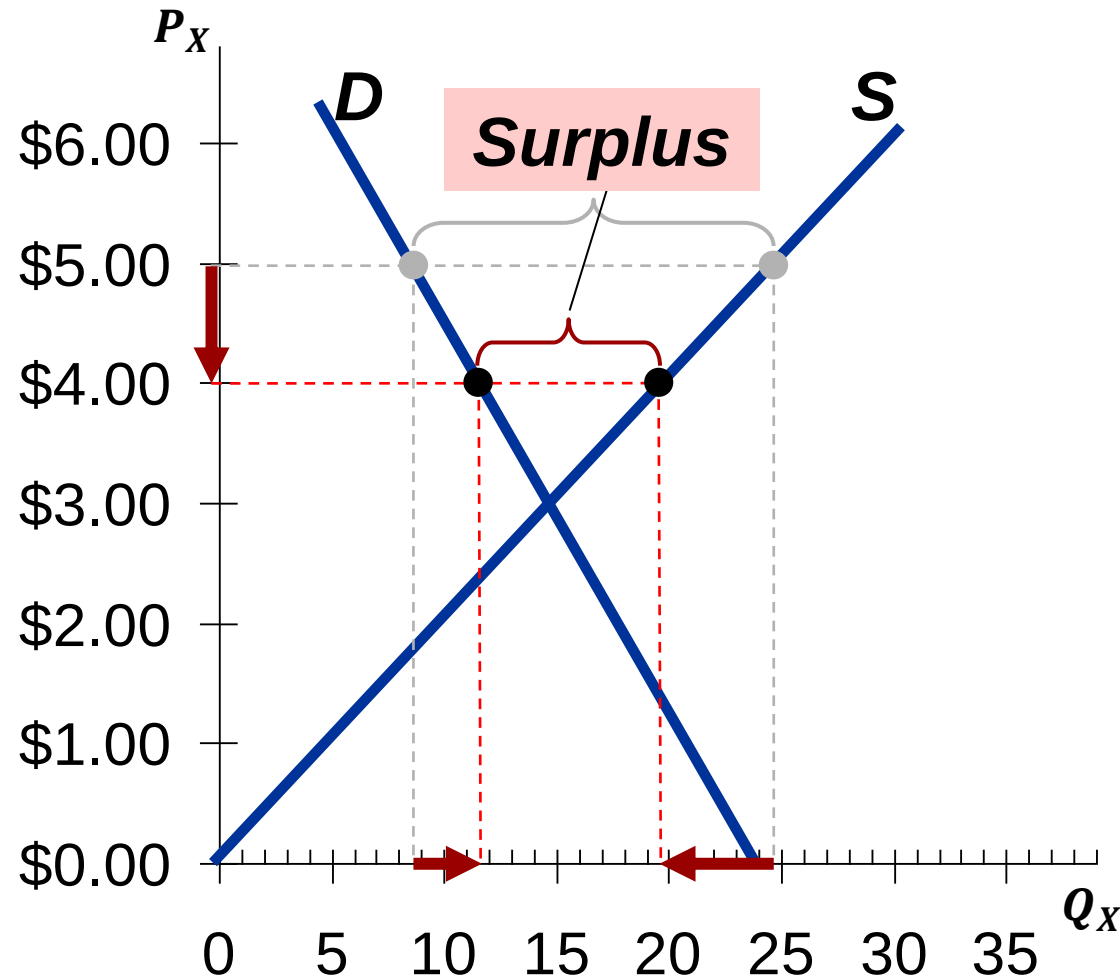


Surplus (excess supply):
quantity supplied is
greater than quantity
demanded

- Example:
if $P = \$5$,
then $Q^D = 9$ lattes
and $Q^S = 25$ lattes

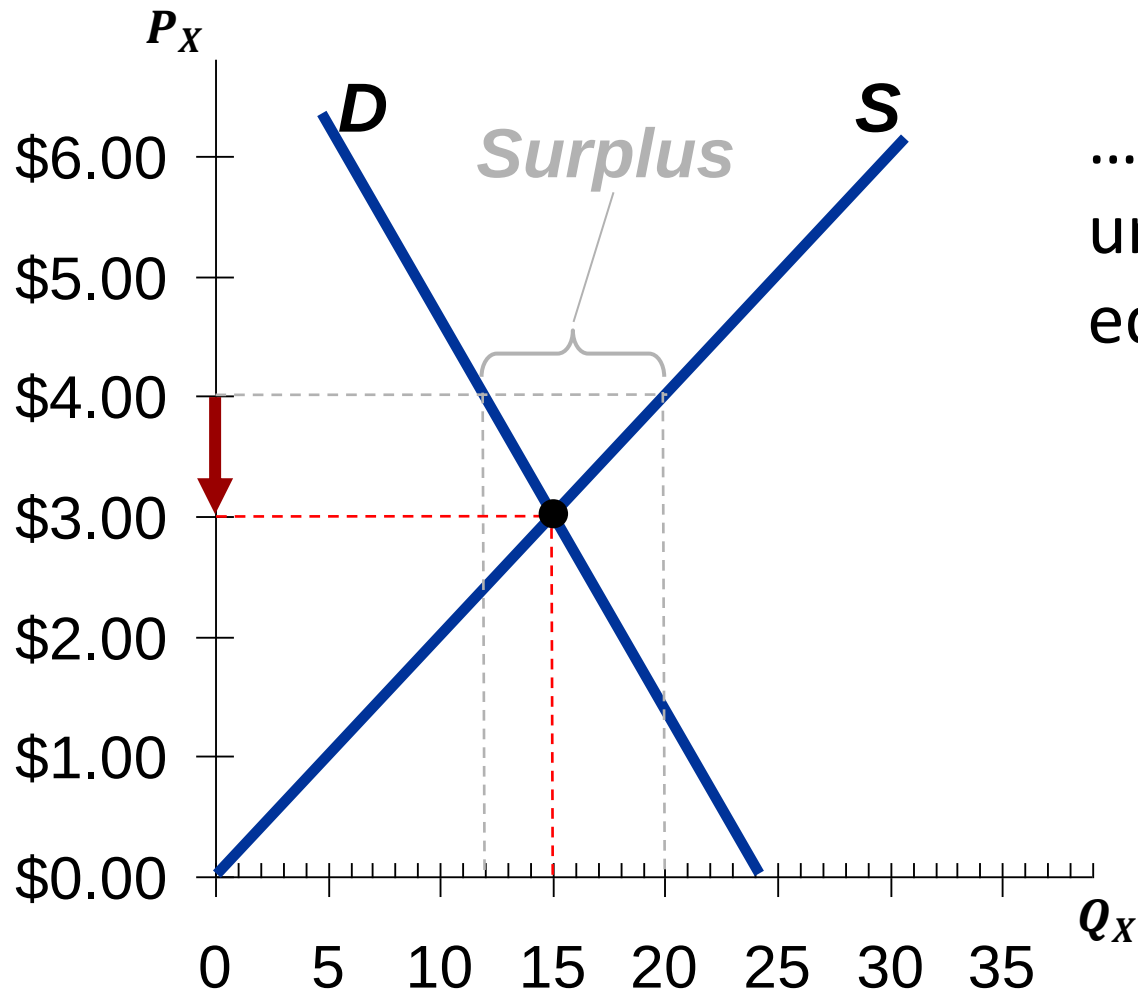
resulting in a surplus of 16
lattes

Markets Not in Equilibrium: **Surplus**



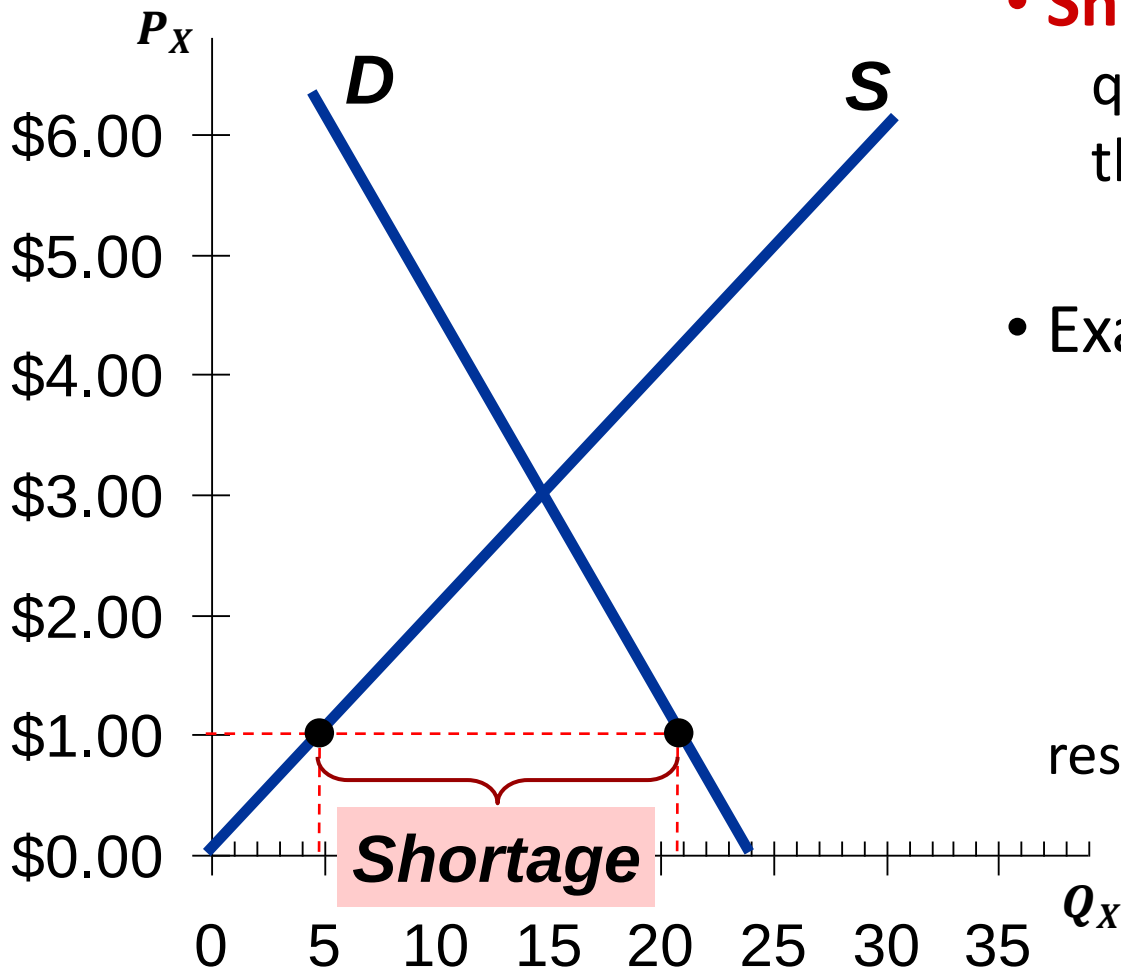
- Facing a surplus, sellers try to increase sales by cutting price.
- This causes Q^D to rise and Q^S to fall. That reduces the surplus.

Markets Not in Equilibrium: **Surplus**



....Prices continue to fall until market reaches equilibrium.

Markets Not in Equilibrium: **Shortage**

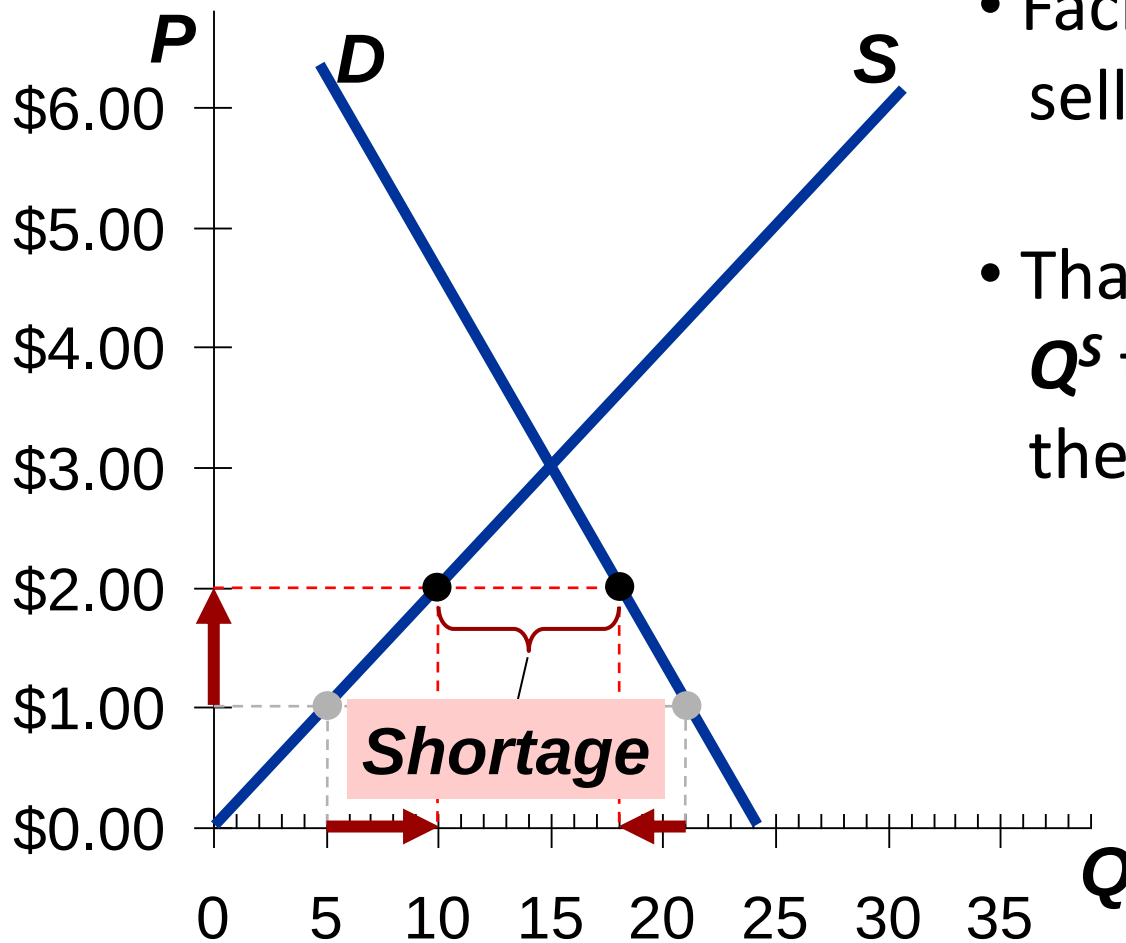


- **Shortage** (excess demand):
quantity demanded is greater
than quantity supplied

- Example:
if $P = \$1$,
then $Q^D = 21$ lattes
and $Q^S = 5$ lattes

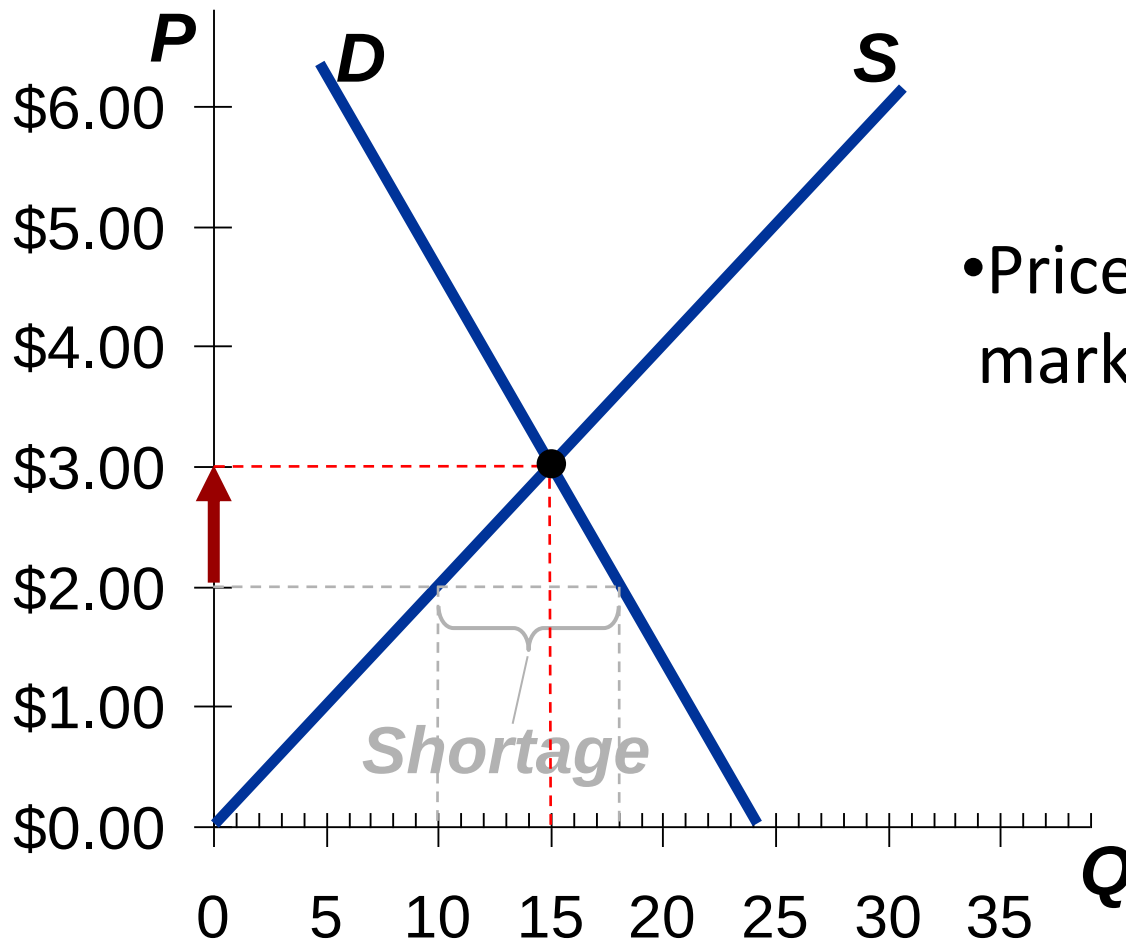
resulting in a shortage of 16 lattes

Markets Not in Equilibrium: **Shortage**



- Facing a shortage, sellers raise the price,
- That causes Q^D to fall and Q^S to rise. That reduces the shortage.

Markets Not in Equilibrium: **Shortage**



- Prices continue to rise until market reaches equilibrium.

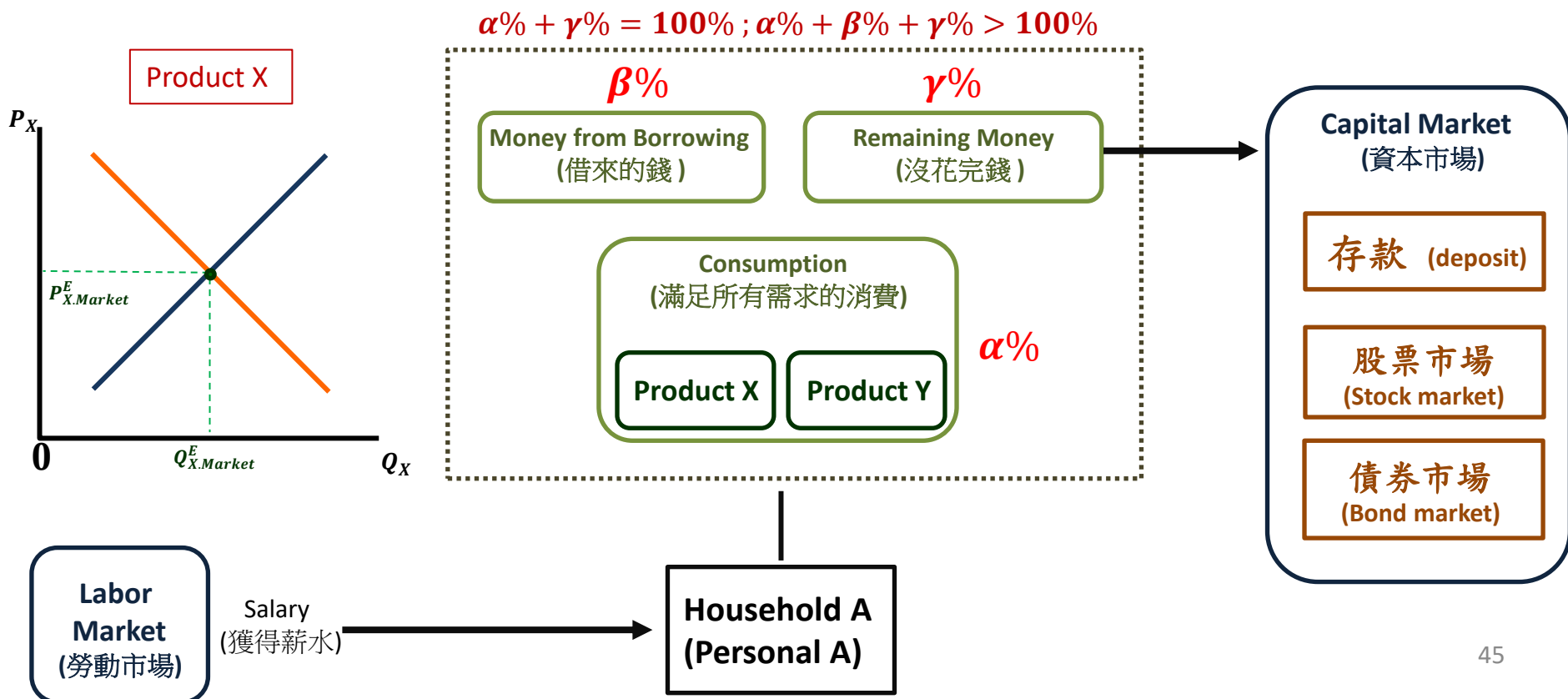
市場機能與資源配置

(Market Mechanism and Resource Allocation)

(以某一家庭為例) Take the decision made by a household for example

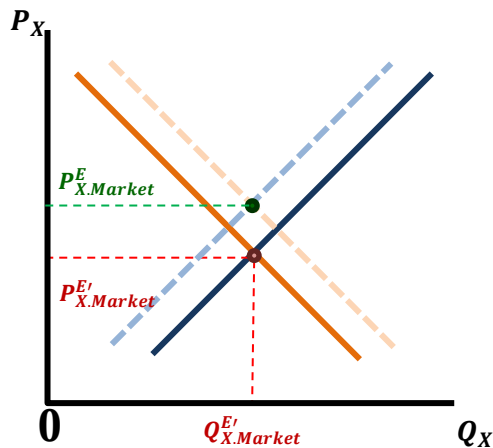
- 當 (X商品) 均衡市價決定之後，市價反過來就成為家庭或廠商決策的重要參考，進而引導資源配置

(As the price of the product X is determined by the market supply and demand, the price will on the other hand influence households' and suppliers' decisions on resource allocation)



$$\alpha^{-}\% + \gamma^{+}\% = 100\%; \alpha^{-}\% + \beta\% + \gamma^{+}\% > 100\%$$

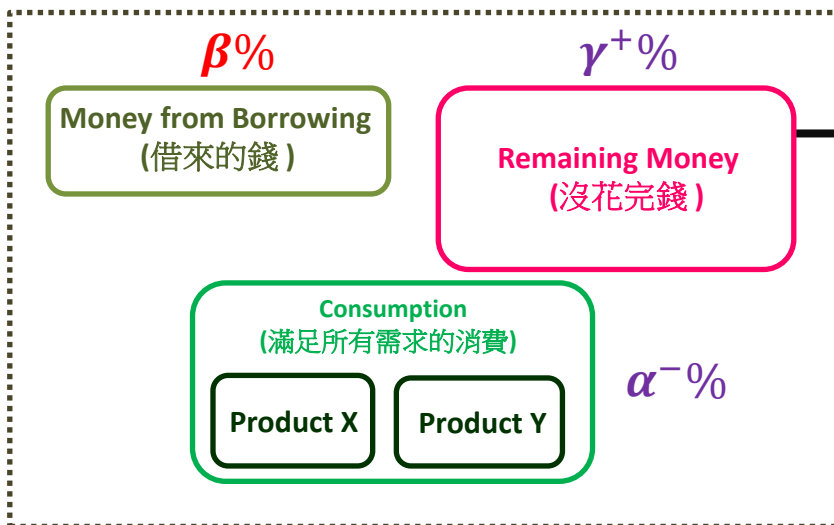
the market price of Product X decreases



Labor Market
(勞動市場)

Salary
(獲得薪水)

Household A
(Personal A)



Capital Market
(資本市場)

存款
(deposit)

股票市場
(stock market)

債券市場
(bond market)

- 當X商品的市場價格下降，且家庭A對X商品的需求量依然固定時（或變動極小），則假定獲得的薪水不變，沒有花完的錢將變多，投入資本市場的錢將隨之變多
- 當有更多的資金湧入資本市場時，將會有更多的公司可以募得資金，讓這些公司用來幫員工加薪，或者是做新的投資

How Prices Allocate Resources

- “Markets are usually a good way to organize economic activity”
- In market economies
 - Prices adjust to balance supply and demand
- These equilibrium prices
 - Are the signals that guide economic decisions and thereby allocate scarce resources

Another interpretation for demand

- If the quantity demanded for product X can be expressed as a demand function:

$$Q_X^D = f(P_X | P_Y, I, R...),$$

this function means that given other things, e.g., $(P_Y, I, R...)$, being equal, the quantity demanded for product X would be Q_X^D if its price is P_X .

The function f could be in the form of

$$Q_X^D = 250 - 0.01 P_X + 0.07 P_Y + 0.02 I - 0.38 R + \dots$$

- The inverse function of f , f^{-1} , could be expressed as

$$P_X = f^{-1}(Q_X^D | P_Y, I, R...).$$

Given other things being equal, if the quantity demanded is Q_X^D , P_X here means the “**maximal price per unit of product X**” the buyers are willing to pay. f^{-1} could be in the form of

$$P_X = 25000 - 100 Q_X^D + 7 P_Y + 2 I - 38 R + \dots$$

Another interpretation for supply

- If the quantity supplied for product X can be expressed as a supply function:

$$Q_X^S = g(P_X | P_Y, P_F, R...),$$

this function means that given other things, e.g., $(P_Y, P_F, R...)$, being equal, the quantity supplied of product X would be Q_X^S if its price is P_X .

The function g could be in the form of

$$Q_X^S = 10 + 0.1 P_X + 0.07 P_Y - 0.35 P_F - 1.33 R + \dots$$

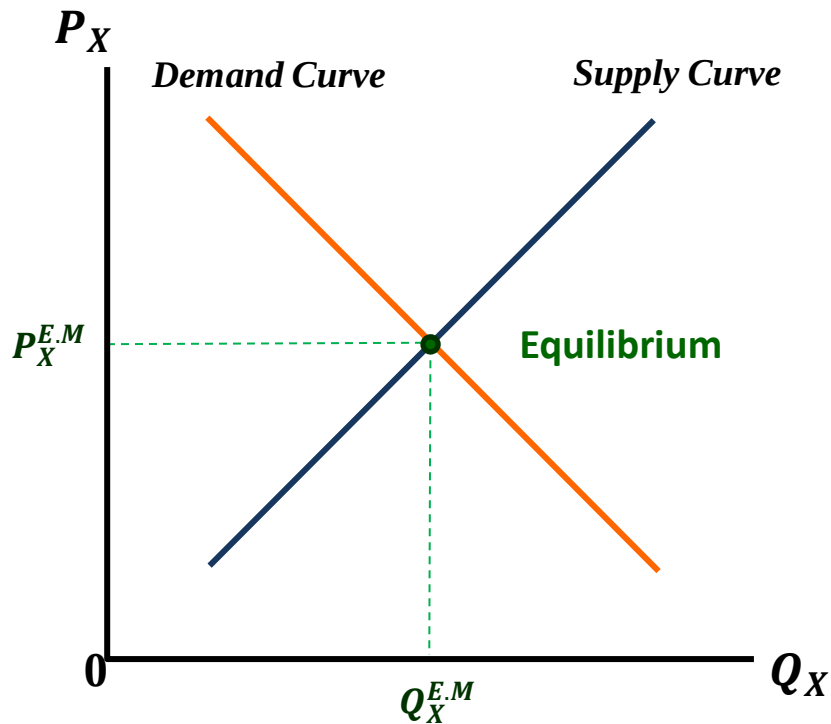
- The inverse function of g , g^{-1} , could be expressed as

$$P_X = g^{-1}(Q_X^S | P_Y, P_F, R...).$$

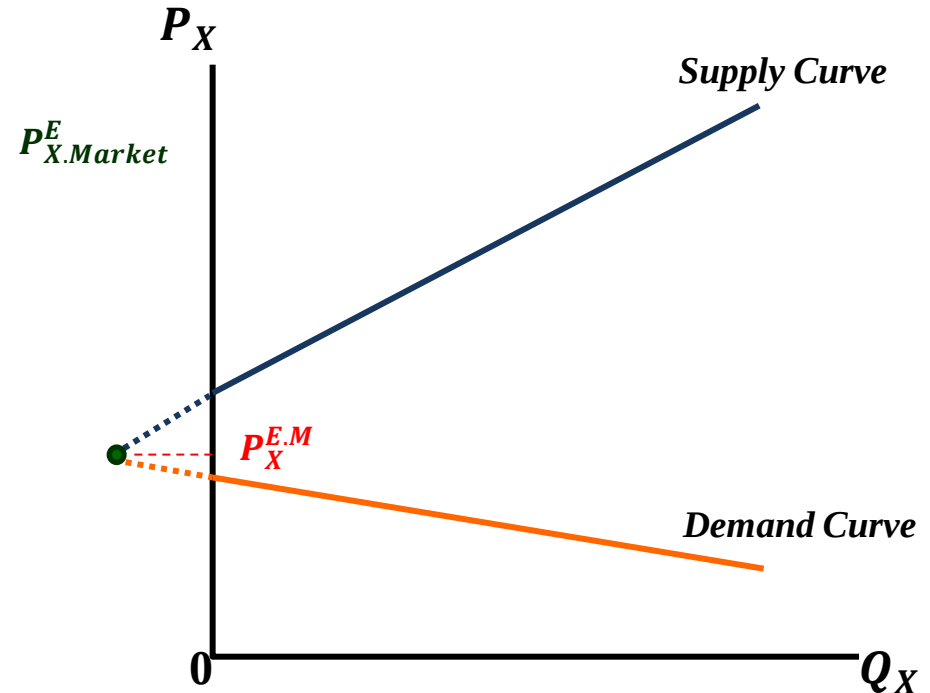
Given other things being equal, if the quantity supplied is Q_X^S , P_X here means the “**minimal price per unit of product X**” the supplier is willing to offer. g^{-1} could be in the form of

$$P_X = -100 + 10 Q_X^S - 0.7 P_Y + 3.5 P_F + 13.3 R + \dots$$

Have quotations, but no transactions



If $Q_X^{E.M} > 0$, that means the product X has quotations and transactions. In other word, that means the maximal price the buyers accept and the minimal price the suppliers accept intersect at $P_X^{E.M}$.



If the product X has quotation $P_X^{E.M}$, but the corresponding $Q_X^{E.M} < 0$, that means the product has quotations but no transactions. The minimal price the suppliers accept is greater than the maximal price the buyers accept.

(最高)願付價格與(最低)願受價格的例子 - 台積電最佳五檔買、賣價

2018/3/2 台積電股票的當日最後成交價(收盤價): 240.00

收盤價時台積電股票的最佳五檔買賣價:

成交之外，買方的最高願付價格

委買價格	委買量 (張)
239.50	432
239.00	1436
238.50	1970
238.00	4280
237.50	1129

賣方的最低願受價格

委賣價格	委賣量 (張)
240.00	1168
240.50	315
241.00	403
241.50	456
242.00	590

- 這代表，當下台積電的股票要能成交，買方的委買價格必須至少要高到240.00元，或賣方的委賣價格必須要低至239.50
- 若趕在最後一筆交易撮合前，賣方出現兩張，以每張239.50元的委托賣出，則最後台積電的收盤價會變成239.50，最高願委買價格所對應的委買量會變成430張

但這會不會是有行無市(有供給但無需求)呢?
(have quotations but no demands?)



古董、古錢、郵票、玉珮
(antiques)

那有市無行呢?(有需求卻無供給)

(could have demands but no quotations?)

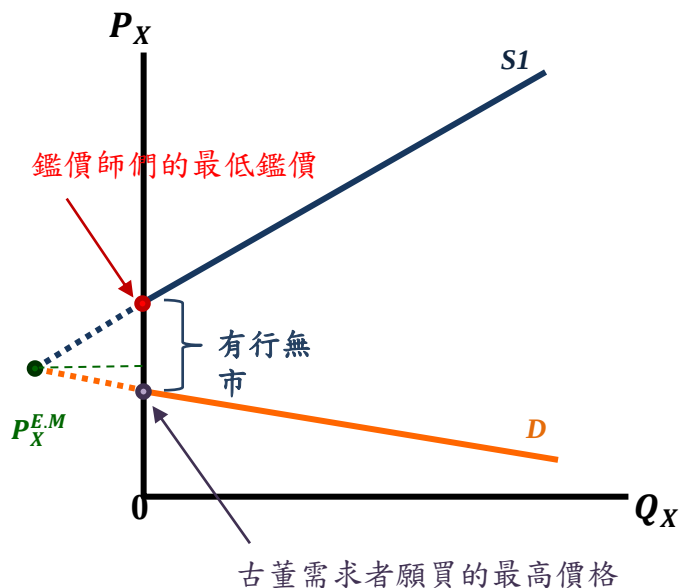
<https://www.youtube.com/watch?v=33Raqx9sFbo>



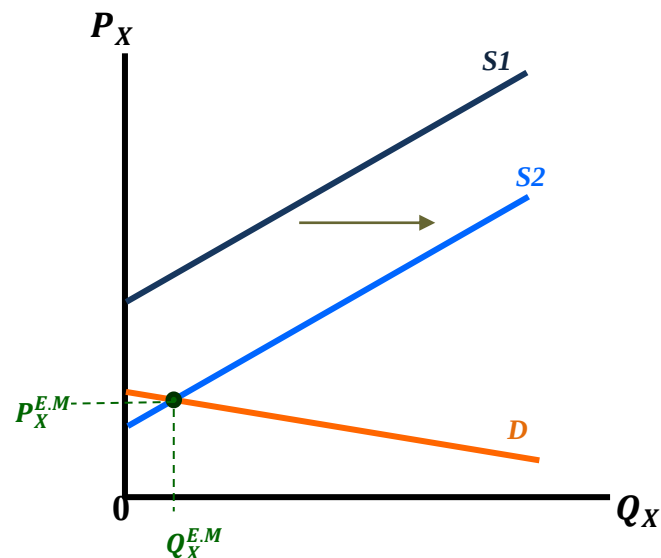
關鍵報告電影上映日: 2002; iphone 第一代: 2007

尖端科技
(cutting-edge technology)

有行無市的古董鑑價、買賣

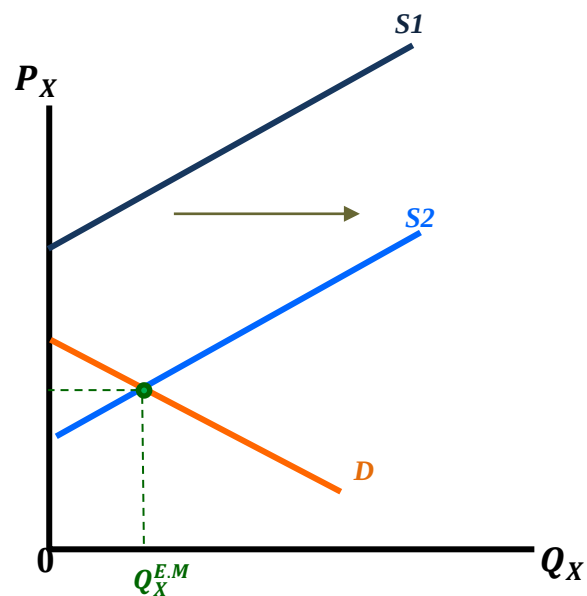
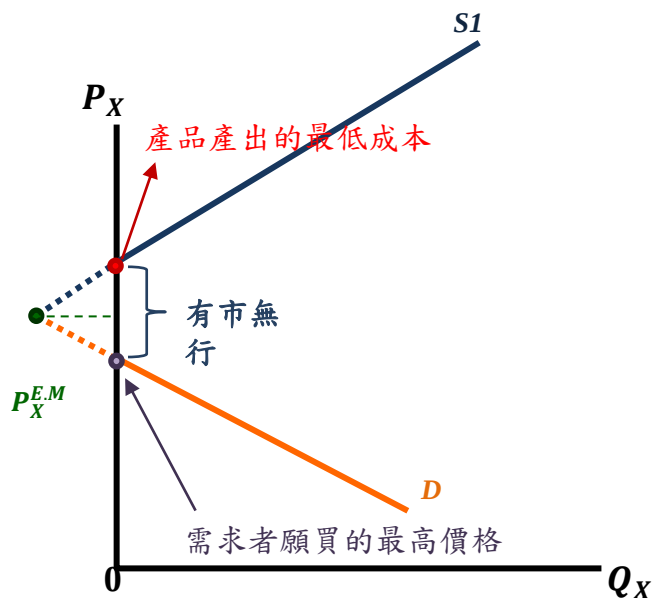


- 誰是古董的需求者：當舖、收藏家（人數少）
- 當鑑價師（最低）開價太高，會沒有需求（ $Q_X^{E.M} < 0$ ）



- 當有更多相同朝代的古物出土，市場上某類古物供應量大增，供給曲線右移，鑑價師的最低鑑價下降
- 低開價，開始有收藏家願意接受，開始有需求（ $Q_X^{E.M} > 0$ ）

有市無行的尖端科技應用-觸控螢幕



- 誰會是觸控螢幕的需求者：手機、電腦使用者（人數超多）
- 但生產成本太高，以至於即便大家有需求，但仍然沒辦法接受（ $Q_X^{E.M} < 0$ ）
- 當技術發展到可以大幅降低生產成本時，則其他條件相同，供給曲線會右移
- 低產品售價，開始有消費者願意接受（ $Q_X^{E.M} > 0$ ）